MILITARY SPECIFICATION

PERISCOPE, TANK: M35E1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of combination passive - daylight tank gunner's periscope designated Periscope, Tank: M35E1. The periscope is a part of the fire control system in the 105MM Gun Tank, M60 and is used for day and night sighting and observing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-E-48441	Elbow Assembly; Passive: 11733401
MIL-B-48447	Body (Daylight) Assembly: 11744900
MIL-H-48590	Head Assembly: 11727486

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U.S. Army Frankford Arsenal

F8247391	Test Fixture for Periscopes M32 and M35
F11727450	Periscope, Tank: M35E1
F11732155	Gage, Interchangeability, Minimum
F11732156	Gage, Interchangeability, Maximum
F11738140	Final Acceptance Inspection Equipment

PACKAGING DATA SHEET

PS11727450	Packaging of Periscope, Tank: M35E1

(Copies of specifications, standards, drawings and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with Drawing F11727450 and drawings pertaining thereto. The following shall have met their respective specifications (except that they need not be subjected to separate temperature, vibration and shock tests) and be interchangeable within the requirements of this specification.

Elbow Assembly, Passive:	11733401	MIL-E-48441
Body (Daylight) Assembly:	11744900	MIL-B-48447
Head Assembly:	11727486	MIL-H-48590

3.2 Interchangeability.

a. Modules.- The periscope shall meet the requirements of 3.6.2, 3.6.5, 3.6.6, 3.6.6.1 and 3.6.8 without need for realignment of components or selection of modules listed in 3.1. Adjustments, other than those specified herein, are prohibited.

b. Periscope with mount.- The periscope shall be capable of insertion into the Mount Periscope: M118E1 (gages are specified in lieu of mount in 4.6.5.2). After insertion, the periscope shall be capable of being firmly secured and subsequently removed.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

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3.4 Environmental.

3.4.1 Storage temperature.- The periscope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperature, plus 60° to plus 90° Fahrenheit (F), after having been exposed and thermally stabilized at temperatures of minus 80° and plus 160°F.

3.4.2 Operating temperatures.- The periscope shall meet the applicable requirements of 3.6.10 while exposed and thermally stabilized at temperatures of minus 40° and plus 150°F.

3.4.3 Vibration/Shock.

3.4.3.1 Vibration "A".- This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401). The vibration input to the periscope shall be at the interface of the periscope and mount. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .65 G (peak).

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .65 to 1.5 G's (peak).

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's (peak).

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G's (peak).

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .65 G (peak).

f. At 5 Hz, the acceleration shall be .65 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test the periscope shall meet the requirements of 3.6.3.2, 3.6.4.1, 3.6.9, 3.6.2, 3.6.3.1, 3.6.3.1.1, 3.6.3.1.2, 3.6.5, 3.6.6, 3.6.6.1, 3.6.7, 3.6.8, 3.2, 3.1, 3.5 and then each module (see 3.1) shall meet their own respective post-vibration test requirements.

3.4.3.2 Vibration "B".- This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401). The vibration input to the periscope shall at the interface

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of the periscope and mount. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G (peak).

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G (peak).

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G (peak).

d. At 5 Hz, the acceleration shall be 1.0 G (peak).

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minute ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). There shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp during vibration and the periscope shall show no evidence of physical failure subsequent to vibration. Following the vibration test the periscope shall meet the requirements of 3.6.3.2, 3.6.4.1, 3.6.9, 3.6.2, 3.6.3.1, 3.6.3.1.1, 3.6.3.1.2, 3.6.5, 3.6.6, 3.6.6.1, 3.6.7, 3.6.8, 3.2, 3.1, 3.5 and then each module (see 3.1) shall meet their own respective post-vibration test requirements.

3.4.3.3 Shock.- This requirement shall be met with the image intensifier tube and reticle projector lamp energized by the application of +24 ± 6.0 VDC to Contact 11733497 (Reference Drawing F11733401). The shock input to the periscope shall be at the interface of the periscope and mount. The periscope shall be subjected to 24 cumulative shocks consisting of three each at 100 G's, then 40 G's, applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The periscope shall also be subjected to six cumulative shocks consisting of three each at 90 G's, then 35 G's applied in one direction along the verticle axis (normal operating position of periscope only). The shock pulse shall have a half sine wave. For 100 G's ± 10% and 90 G"s ± 10% the shock pulse shall have a time duration of 3 ± .2 milliseconds; for 40 G's ± 10% and 35 G's ± 10% the shock pulse shall have a time duration of 18 ± 3 milliseconds. Subsequent to shock there shall be no failure or flicker of light emission from the image tube nor the reticle projector lamp. The periscope shall also show no evidence of physical failure. Following the shock test the periscope shall meet the requirements of 3.6.3.2, 3.6.4.1, 3.6.9, 3.6.2, 3.6.3.1, 3.6.3.1.1, 3.6.3.1.2, 3.6.5, 3.6.6, 3.6.6.1, 3.6.7, 3.6.8, 3.2, 3.2, 3.1, 3.5 and then each module (see 3.1) shall meet their own respective post-shock test requirements.

3.5 Cleanliness.- With the periscope disassembled, the three modular units shall meet the following requirements:

a. Defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end of each module (including the unity power system).

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b. Dirt.

(1) Body assembly (Daylight). There shall be no more than three (3) particles of dirt or foreign material (classified as dirt) on any objective or eyepiece assembly. The apparent size of any dirt particle shall not exceed 0.05 inch.

(2) Elbow assembly (passive). When viewing through the eyepiece with the image intensifier tube energized, there shall be no more than five (5) particles visible on the screen surface. The size of any particle shall not exceed the apparent width of the reticle line.

(3) Head assembly. There shall be no more than three (3) particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05 inch.

3.6 Performance.

3.6.1 Orientation.

a. Datum Plane "W" (Ref. Dwg. F11747303) defined by the four mounting pads in a vertical plane. Datum Plane "J" (Ref. Dwg. F11727486) on a horizontal plane.

b. Side "L" (Ref. Dwg. F11727486) of the input coupling positioned 22 degrees 30 minutes with respect to Datum Plane "W" as measured in a vertical plane.

NOTE: At no time shall the input coupling be allowed to strike hard against the travel stops.

c. The requirements of 3.6.3.1 to 3.6.3.2 inclusive, shall be met with:

(1) The clear filter positioned in the field of view.
(2) An infinity target which is projected into Lens 8289292 perpendicular to Datum Plane "W"
(3) The eyepiece of the daylight system focused on the image of the projected infinity target.
(4) The eyepiece of the Passive System focused on the screen of the image intensifier tube.
(5) A 1200 meter target provided for both the Daylight and Passive Systems

d. Focus passive elbow assembly by rotating objective focusing Ring 11733436 for best resolution of target (1200 meters) image.

3.6.2 Parallax.- Parallax between the reticle and the image of a target at 1200 meters shall not exceed 0.1 mil when adjusted by Eccentric 10516549 on the daylight body assembly.

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3.6.3 Collimation.- The following collimation requirements shall be established prior to meeting the vibration/shock requirements of 3.4.3.

a. An incident line of sight which is originally perpendicular within 0.2 mil to Datum Plane "W".

b. The input coupling positioned in accordance with 3.6.1b.

3.6.3.1 Pre-Vibration/Pre-Shock, Passive reticle setting.- Rotate passive boresight knobs to establish coincidence of the boresight cross with the incoming line of sight (see 3.6.3a) prior to meeting the requirements of 3.4.3.

3.6.3.1.1 Pre-Vibration/Pre-Shock, daylight viewing area.- An incident line of sight which is originally perpendicular to Datum Plane "W" shall coincide within 0.2 mil, with the image of an infinity target which is projected into Lens 8289292 perpendicular to Datum Plane "W" prior to meeting the requirements of 3.4.3.

3.6.3.1.2 Passive viewing area.

a. Elevation knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in elevation and depression from the collimated setting of 3.6.3a.

b. Deflection knob travel.- The boresight knob shall provide a minimum travel of 6.5 mils in each direction from the collimated setting of 3.6.3a.

c. Line of sight variation.- Variation in the line of sight in elevation, as the deflection knob is rotated by at least 6.5 mils in each direction from the collimated setting of 3.6.3a shall not exceed 0.3 mil (total variation not to exceed 0.6 mil).

3.6.3.2 Post-Vibration/Post-Shock collimation.- The original collimation settings established in 3.6.3.1 and 3.6.3.1.1 shall not deviate more than 0.3 mil (including fixture error) as observed through eyepiece subsequent to meeting requirements of 3.4.3.

3.6.4 Collimation repeatability.

3.6.4.1 Assembling daylight body and passive elbow to head assembly.- The daylight body and passive elbow assemblies shall be capable of repeating an original line of sight within 0.20 mil when removed and reinstalled (a minimum of 3 times) into the head assembly.

3.6.5 Image tilt - daylight and passive.- The image of a target line which is originally parallel to Datum Plane "J" shall emerge from the eyepiece parallel to this same surface within 2.0 degrees of arc.

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3.6.6. Parallelism of reticle and image, passive.- The image of a target line originally parallel to Datum Plane "J" shall be parallel to the horizontal reticle line within 2.0 degrees of arc.

3.6.6.1 Parallelism of target image and projected image.- The image of a target line which is originally parallel to Datum Plane "J" and perpendicular to Datum Plane "W" shall emerge from the eyepiece parallel to Datum Plane "J" within 2.0 degrees of arc to the image of an infinity target line projected into Lens 8289292 which is originally parallel to Datum Plane "J" and perpendicular to Datum Plane "W".

3.6.7 Objective focusing range.

3.6.7.1 Daylight.- Rotation of Eccentric 10516549 shall cause the objective assembly to travel a minimum of 0.010 inch in each direction from the position established in 3.6.2.

3.6.7.2 Passive.- Rotation of Ring 11733436 shall cause the objective to bring targets from 50 meters to 3000 meters into focus.

3.6.8 Resolution.

3.6.8.1 Daylight.- The resolution shall be 10 seconds or arc or less in all four meridians. Eyepiece focus shift between any two meridians shall not exceed 0.25 diopter.

3.6.8.2 Passive.- The resolution shall be 1 minute of arc or less in both meridians. Eyepiece focus shift between the meridians shall not exceed 0.25 diopter.

3.6.9 Electrical characteristics - passive.

3.6.9.1 Image intensifier tube.- The image intensifier tube shall be energized by the application of +24 $\pm$ 6.0 VDC to the Contact 11733497 or application of 3 VDC $\pm$ 0.3 VDC to Connector 11733493 (Ref. Dwg. F11733401).

3.6.9.2 Image steadiness.- There shall be no visible flicker, no movement of the image nor audible sparking when observed for 1 minute. This observation shall be made after the expiration of 15 seconds tube stabilization time.

3.6.9.3 Reticle brightness.- With a minimum of 18 and a maximum of 30 volts applied to the Contact 11733497 the reticle markings shall be visible when viewing through the eyepiece with tthe image intensifier tube energized (see 3.6.9.1). The reticle brightness shall be adjustable from off to full bright by adjusting the reticle brightness control Knob MS91528-0E1B clockwise.

3.6.9.4 On-Off Switch.- The image intensifier tube shall shut down when + 24 VDC $\pm$ 6 VDC is removed from Contact 11733497 with the on-off switch in the on position. The tube shall not be energized by reapplication of the 24 VDC unless the on-off switch is recycled from off to on.

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3.6.10 Operability.

3.6.10.1 Elevation input coupling.- The running torque required to rotate the elevation input coupling through maximum elevation and depression, shall not exceed six pound-inches (lb-ins) at standard ambient temperature or 10 lb-ins at extreme temperatures of minus 40° and plus 150°F.

3.6.10.2 Diopter movement - Daylight and Passive.- The running torque required to rotate the daylight diopter movement through a minimum range of minus 3.0 to plus 3.0 diopters and the passive diopter movement through a minimum range of minus 4.0 to plus 4.0 diopters shall be within the limits of 1.5 and 6.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 15 lb-ins.

3.6.10.3 Focusing mechanism - Passive.- The running torque required to rotate Ring 11733436 through the range specified in 3.8.7 shall be within the limits of 4.0 and 14.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 24.0 lb-ins.

3.6.10.4 Disengagement of boresight knob locks Passive.- The direct force required to disengage each boresight knob from the lock position shall be within the limits of 2.0 to 5.0 pounds (lbs) when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the direct force shall not exceed 9.0 lbs.

3.6.10.5 Rotation of boresight knobs Passive.- The running torque required to rotate each boresight knob through the travel range specified in 3.6.3.1.2a and 3.6.3.1.2b respectively, shall be within the limits of 0.5 and 5.0 lb-ins when checked at standard ambient temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 8.0 lb-ins.

3.6.10.6 Shutter lever and on-off switch - Passive.- The torque required to rotate the shutter lever from the off to on position shall be within the limits of 4.5 to 6.5 lb-ins when checked at standard ambient temperature. When checked at temperature conditions specified in 3.4.2, the torque shall not exceed 10 lb-ins.

3.6.10.7 Gain adjust and reticle brightness controll - Passive.- This requirement shall be met at standard ambient temperature and the temperature conditions specified in 3.4.2. Knob MS 91528-0E1B shall be manually operable through the full limits of the rheostat.

3.6.10.8 Focusing lock - Daylight.- This requirement shall be met with a 10 lb-ins torque applied to the locking screw on Clamp 10516536. When checked at standard ambient temperature and the temperature conditions specified in 3.4.2, there shall be no rotation of Eccentric 10516549 when a 48 ounce-inches (oz-ins) torque is applied to the eccentric.

3.6.10.9 Objective focusing mechanism - Daylight.- This requirement shall be met with the locking screw backed off a minimum of two full revolutions from the locked position established in 3.6.10.8. The running torque

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required to rotate Eccentric 10516549 through a minimum of one full revolution in each direction shall not exceed 30 oz-ins when checked at standard temperature. When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 50 oz-ins.

3.6.10.10 Filter assembly - Daylight.- This requirement shall be met at standard temperature and the temperature conditions specified in 3.4.2. Handle 10516536 shall be manually operable to position each of the four filters in the field of view.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three periscopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

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4.3.1.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.1.1 Periscopes.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III, IV, and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements and related tests shall be subjected to 100% inspection:

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
1.	Line of sight variation, passive viewing area	3.6.3.1.2c	4.6.4.3.3
2.	Resolution, daylight viewing area	3.6.8.1	4.6.4.8
3.	Resolution, passive viewing area	3.6.8.2	4.6.4.9
4.	Image intensifier tube	3.6.9.1	4.6.4.1.1
5.	Image steadiness	3.6.9.2	4.6.4.1.2
6.	Reticle Brightness	3.6.9.3	4.6.4.1.3
7.	On-Off switch	3.6.9.4	4.6.4.1.4
8.	Interchangeability, periscope with mount	3.2b	4.6.5.2

NOTE: The tests in 4.3.2.2 shall be conducted at a temperature between +60° and +90°F.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS
CRITICAL: NONE
MAJOR: AQL 0.65% Defective

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Pre-vibration collimation, passive reticle setting	3.6.3.1	4.6.3.1
102.	Pre-vibration collimation, daylight viewing area	3.6.3.1.1	4.6.3.1.1
103.	Vibration "B"	3.4.3.2	4.6.3.3

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104.	Post-vibration collimation, passive reticle setting	3.6.3.2	4.6.3.5 & 4.6.3.5.1
105.	Post-vibration collimation, daylight viewing area	3.6.3.2	4.6.3.5 & 4.6.3.5.2
106.	Cleanliness	3.5	4.6.3.8
107.	Parallax	3.6.2	4.6.4.2 & 4.6.4.2.1
108.	Elevation knob travel, passive viewing area	3.6.3.1.2a	4.6.4.3.1
109.	Deflection knob travel, passive viewing area	3.6.3.1.2b	4.6.4.3.2
110.	Collimation repeatability, assembling daylight body to head assembly	3.6.4.1	4.6.3.6.1
111.	Collimation repeatability, assembling passive elbow to head assembly	3.6.4.1	4.6.3.6.2
112.	Image tilt, daylight and passive	3.6.5	4.6.4.4
113.	Parallelism of reticle and image, passive	3.6.6	4.6.4.5
114.	Parallelism of target image and projected image	3.6.6.1	4.6.4.6
115.	Objective focusing range, daylight	3.6.7.1	4.6.4.7.1
116.	Objective focusing range, passive	3.6.7.2	4.6.4.7.2
117.	Operability, disengagement of boresight knob locks passive	3.6.10.1	4.6.2.1
118.	Operability, diopter movement daylight and Passive	3.6.10.2	4.6.2.2
119.	Operability, focusing mechanism -Passive	3.6.10.3	4.6.2.3
120.	Operability, disengagement of boresight knob locks- Passive	3.6.10.4	4.6.2.4
121.	Operability, rotation of boresight knobs Passive	3.6.10.5	4.6.2.5
122.	Operability, shutter lever and on-off switch - passive	3.6.10.6	4.6.2.6
123.	Operability, gain adjust and reticle brightness control Passive	3.6.10.7	4.6.2.7
124.	Operability, focusing lock Daylight	3.6.10.8	4.6.2.8
125.	Operability, objective focusing mechanism - Daylight	3.6.10.9	4.6.2.9
126.	Operability, filter assembly	3.6.10.10	4.6.2.10

MINOR: NONE

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NOTE: The tests in Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One periscope shall be selected at random as a special sampling from each 100 produced. the sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings and specifications - visual
302.	General specification	3.3	MIL-F-13926 - visual

4.4.1.1 Three periscopes that have met the requirements and tests in 4.3.2.2 and in Table I shall be selected at random as a special sample from each months production. Each sample shall then meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Interchangeability, modules	3.2a	4.6.5.1

4.4.1.2 Environmental.- Three periscopes shall be selected at random as samples from each 50 produced, or from each months production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in 4.3.2.2 and in Table I.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (-80° and +160°F)	3.4.1	4.6.1.1
305.	Operating temperatures (-40° and +150°F)	3.4.2	4.6.1.2
306.	Operability, elevation input coupling	3.4.2, 3.6.10.1	4.6.1.2, 4.6.2.1
307.	Operability, diopter movement - Daylight and Passive	3.4.2, 3.6.10.2	4.6.1.2, 4.6.2.2
308.	Operability, focusing mechanism Passive	3.4.2, 3.6.10.3	4.6.1.2, 4.6.2.3

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309.	Operability, disengagement of boresight knob locks - Passive	3.4.2, 3.6.10.4	4.6.1.2, 4.6.2.4
310.	Operability, rotation of boresight knobs Passive	3.4.2, 3.6.10.5	4.6.1.2, 4.6.2.5
311.	Operability, shutter lever and on-off switch - Passive	3.4.2, 3.6.10.6	4.6.1.2, 4.6.2.6
312.	Operability, gain adjust and reticle brightness control - Passive	3.4.2, 3.6.10.7	4.6.1.2, 4.6.2.7
313.	Operability, focusing lock - Daylight	3.4.2, 3.6.10.8	4.6.1.2, 4.6.2.8
314.	Operability, objective focusing mechanism-Daylight	3.4.2, 3.6.10.9	4.6.1.2, 4.6.2.9
315.	Operability, filter assembly	3.4.2, 3.6.10.10	4.6.1.2, 4.6.2.10

NOTE: The tests for characteristics 306 through 315 inclusive, shall be conducted at -40° and +150°F.

4.4.4 Verification, follow-up.- Three periscopes shall be randomly selected as samples from production lots at 25 percent, 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table V.

TABLE V - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
316.	Pre-vibration collimation, passive reticle setting	3.6.3.1	4.6.3.1
317.	Pre-vibration collimation, daylight viewing area	3.6.3.1.1	4.6.3.1.1
318.	Vibration "A"	3.4.3.1	4.6.3.2
319.	Post-vibration collimation, passive reticle setting	3.6.3.2	4.6.3.5 & 4.6.3.5.1
320.	Post-vibration collimation, daylight viewing area	3.6.3.2	4.6.3.5 & 4.6.3.5.2
321.	Pre-shock collimation, passive reticle setting	3.6.3.1	4.6.3.1
322.	Pre-shock collimation, daylight viewing area	3.6.3.1.1	4.6.3.1.1
323.	Shock	3.4.3.3	4.6.3.4
324	Post-shock collimation, passive reticle setting	3.6.3.2	4.6.3.5 & 4.6.3.5.1
325.	Post-shock collimation, daylight viewing area	3.6.3.2	4.6.3.5 & 4.6.3.5.2

4.4.5 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specification or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Special inspection equipment.

4.5.2.3.1 Performance.- Special testing equipment to test performance of the complete periscope shall conform to the requirements of drawing F11738140 (Test Set-up, Periscope) and all drawings pertaining thereto.

4.6 Test methods and procedures.

NOTE: At no time during the following tests shall the input coupling be allowed to strike hard against the travel stops.

4.6.1 Environmental.

4.6.1.1 Storage temperature.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.1. The periscope shall be placed in the test chamber and the temperature of the chamber reduced gradually (rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling test) to -80°F. The periscope shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber

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shall be gradually raised (see above rate of temperature change) to +160°F. The periscope shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F), and the periscope removed from the test chamber. The periscope shall then be subjected to the visual and audible examination of 4.6.3.5. Periscopes which have passed the above examination shall be subjected to the remaining tests, specified in 4.6.1.2 through 4.6.5.2, inclusive. The periscopes shall conform to the requirements of 3.4.1.

NOTE: Temperature testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to temperature cycling (see 3.1). A record of all data necessary to determine compliance with required performance shall be available.

4.6.1.2 Operating temperatures.- The test equipment utilized in this test shall be accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.4.2. The periscope shall be placed in the test chamber and the temperature of the chamber reduced gradually (rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling test) to -40°F. The periscope shall remain at this temperature for a minimum of five hours. At the completion of this five hour period and while at -40°F, the periscope shall be subjected to the tests in 4.6.2 to determine compliance with the applicable requirements of 3.6.10, for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see above rate of temperature change) to +150°F. The periscope shall remain at this temperature for a period of five hours. At the completion of the five hour period and while at +150°F, the periscope shall again be subjected to the tests in 4.6.2 to determine compliance with the applicable requirements of 3.6.10, for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see above rate of temperature change) to room ambient temperature (+60° to +90°F) and the periscope removed from the test chamber.

NOTE: Temperature testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to temperature cycling (see 3.1). A record of all data necessary to determine compliance with required performance shall be available.

4.6.2 Operability.

4.6.2.1 Elevation input coupling.- Standard torque measuring equipment capable of measuring the torque values specified in 3.6.10.1 shall be utilized for this test. An appropriate adapter to accommodate the coupling shall be used with the torque measuring equipment. The running torque required to rotate the elevation input coupling through its full range of travel in each direction shall not exceed the value specified in 3.6.10.1 for the temperatures indicated.

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4.6.2.2 Diopter movement - Daylight and Passive.- Standard torque measuring equipment capable of measuring the torque values specified in 3.6.10.2 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter movement through the range specified in 3.6.10.2. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.6.10.2 for the temperatures indicated.

4.6.2.3 Focusing mechanism - Passive.- Standard torque measuring equipment capable of measuring the torque values specified in 3.6.10.3 shall be utilized for this test. An appropriate adapter, to accomodate Ring 11733436 shall be used with the torque measuring equipment. Torque shall be measured while rotating the ring through the range specified in 3.6.10.3 The running torque required to rotate Ring 11733436 shall conform to the applicable requirements of 3.6.10.3 for the temperatures indicated.

4.6.2.4 Disengagement of boresight knob locks - Passive.- Standard measuring equipment capable of measuring the force values specified in 3.6.10.4 shall be utilized for this test. The force required to disengage each boresight knob from the lock position shall conform to the applicable requirements of 3.6.10.4 for the temperatures indicated.

4.6.2.5 Rotation of boresight knobs - Passive.- Standard torque measuring equipment capable of measuring the torque values specified in 3.6.10.5 shall be utilized for this test. An appropriate adapter, to accommodate the boresight knobs shall be used with the torque measuring equipment. Torque shall be measured while rotating each boresight knob through the range specified in 3.6.10.5. The running torque required to rotate each boresight knob shall conform to the applicable requirements of 3.6.10.5 for the temperature indicated.

4.6.2.6 Shutter lever and on-off switch - Passive.- Standard torque measuring equipment capable of measuring the torque values specified in 3.6.10.6 shall be utilized for this test. The torque required to rotate the shutter lever from the off to on position and then back to the off position shall conform to the applicable requirements of 3.6.10.6 for the temperatures indicated.

4.6.2.7 Gain adjust and reticle brightness control - Passive.- Manually operate the Knob MS 91528-0E1B. The knob shall meet the requirements specified in 3.6.10.7, at the temperatures indicated in 3.4.2 in addition to the standard ambient temperature (+60° to +90°F).

4.6.2.8 Focusing lock - Daylight.- A torque wrench capable of applying the specified torque values in 3.6.10.8, together with a standard type driver bit that properly fits the screw slot of the locking screw and eccentric, shall be utilized for performing this test. With the specified torque applied to the locking screw and while observing the eccentric, apply, in a clockwise direction, the specified torque to the eccentric. Then, while still observing the eccentric, apply, in a counterclockwise direction, the specified torque to the eccentric. To conform to the requirements of 3.6.10.8, there shall be no rotation of the eccentric when observed during each torque application.

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4.6.2.9 Objective focusing mechanism - Daylight.- Back off the locking screw on clamp 10516556 a minimum of two full revolutions from its locked position. Torque shall be measured while rotating the eccentric through the range specified in 3.6.10.9. The running torque required to rotate the eccentric shall conform to the applicable requirements of 3.6.10.9 for the temperatures indicated. Upon completion of this test, return the eccentric to the position in 3.6.2 and torque the locking screw as specified in 3.6.10.8.

4.6.2.10 Filter assembly.- Manually operate the Handle, 10516536. The handle shall be capable of positioning each of the four filters into the field of view. The handle shall conform to the requirements of 3.6.10.10 for the temperatures indicated in 3.4.2. Periscopes which have passed the above tests of 4.6.2.1 through 4.6.2.10 inclusive, shall be subjected to the tests in 4.6.3.

4.6.3 Vibration/Shock (Periscope).-

4.6.3.1 Pre-vibration/Pre-shock collimation, passive reticle setting.- This test and the following tests of 4.6.3.1.1, 4.6.3.5.1 and 4.6.3.5.2 shall be performed with the periscope positioned and adjusted as in 4.6.4. Prior to performing these tests, disengage the index pin "D" from the index plate of the test fixture and place the 10 second level on surface "C" of the leveling base. The bubble of the level shall be centered within its graduation lines. If it is not, then adjustment is required in accordance with the procedures prescribed in paragraphs 7 and 8 of the set-up instructions contained on Drawing F8247391. With the index pin "D" disengage from the index plate and surface "C" of the leveling base level, view through the periscope eyepiece (passive) and obtain coincidence of the periscope reticle boresight cross and the target image crosslines, by means of the passive boresight knobs (do not slip knob scales). When coincidence has been obtained, record the boresight knob scale settings for use in the test in 4.6.3.5.1. Without disturbing the knob scale settings, the periscope shall be then subjected to the test in 4.6.3.1.1.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original recorded settings throughout remainder tests herein.

4.6.3.1.1 Pre-vibration/Pre-shock collimation, daylight viewing area.- While viewing through the periscope eyepiece (daylight), observe the target (1200 meters) image. The intersecting point of the vertical and horizontal reticle lines of the infinity target image projected into Lens 8289292, shall be superimposed on the intersecting point of the vertical and horizontal lines of the target (1200 meters) image within the tolerance specified in 3.6.3.1.1, as measured on the target. Record the measurement for use in the test in 4.6.3.5.2.

NOTE: Insure that knob scales are not disturbed (re-slipped) from their original recorded settings throughout remainder of tests herein.

4.6.3.2 Vibration "A".- Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 and 4.6.3.1.1 to determine compliance with 3.6.3. Subsequent to the above tests, the periscope shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and

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and frequency range shall be as specified in 3.4.3.1. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the periscope along three mutually perpendicular axes while the periscope is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11727450). The periscope shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during vibration. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with 3.4.3.1. After allowing 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (passive). The periscope shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.4.3.1. During each vibration sweep cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration test the periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.6.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.4, 4.6.5, 4.6.3.7 and 4.6.3.8 respectively, to determine overall conformance with 3.4.3.1.

NOTE: Vibration testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to vibration (see 3.1). A record of all data (collimation, pre-vibration settings, etc.) necessary to determine compliance with required performance shall be available.

4.6.3.3 Vibration "B".- Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 and 4.6.3.1.1 to determine compliance with 3.6.3. Subsequent to the above tests, the periscope shall be vibrated in accordance with Test Method 514.2, Procedure VIII (Tracked Vehicles) of MIL-STD-810, except the test level and frequency range shall be as specified in 3.4.3.2. Vibration shall be applied along each of three mutually perpendicular axes of the periscope. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the "Tolerance of Test Conditions" specified in the "General Requirements" of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the periscope along three mutually perpendicular axes while the periscope is maintained in it"s normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing F11727450). The periscope shall be clamped on the vibrating equipment by means of an

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appropriate adapter having the same interface features as Mount Periscope: M118E1. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during vibration. Prior to vibration, the image intensifier tube and the reticle projector lamp shall be energized in accordance with 3.4.3.2. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece. The periscope shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.4.3.2. During each vibration sweep cycle, the screen surface of the tube and the reticle pattern shall be monitored to assure that there is no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the vibration test, the periscope shall be subjected to the test. The periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.6.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.4, 4.6.5, 4.6.3.7 and 4.6.3.8 respectively, to determine overall conformance with 3.4.3.2.

NOTE: Vibration testing of each head, daylight body and passive elbow assembly is not required when the complete periscope is subjected to vibration (see 3.1). A record of all data (collimation, pre-vibration settings, etc.) necessary to determine compliance with required performance shall be available.

4.6.3.4 Shock.- Prior to performing this test, the periscope shall be subjected to the test procedure specified in 4.6.3.1 and 4.6.3.1.1 to determine compliance with 3.6.3. The shock equipment and test procedure shall be in accordance with Procedure I, of Method 516.2 specified in MIL-STD-810, except conditions such as "G" values, time duration and wave form shock pulse shall be as specified in 3.4.3.3. The periscope shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. Prior to shock, the image intensifier tube and reticle projector lamp shall be energized in accordance with 3.4.3.3. The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops during the shock test. After allowing a 15 second tube stabilization period, the screen surface of the tube shall be illuminated and the reticle pattern shall be projected into the optical system when observed through the periscope eyepiece (passive). The periscope shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position of the periscope). Subsequent to each shock, the screen surface of the tube and the reticle pattern shall be observed. During these observations, there shall be no failure or flicker of light emission from the image intensifier tube nor the reticle projector lamp. Immediately following the shock test, the periscope shall be subjected to the tests in 4.6.3.5 through 4.6.3.6.2 inclusive. Periscopes that have passed the above tests shall then be subjected to the tests specified in 4.6.4, 4.6.5, 4.6.3.7 and 4.6.2.8 respectively, to determine overall conformance with 3.4.3.3.

NOTE: Shock testing of each head, daylight body and passive elbow assembly is not required when the complete

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periscope is subjected to shock (see 3.1). A record of all data (collimation, pre-shock settings, etc.) necessary to determine compliance with required performance shall be available.

4.6.3.5 Post-vibration/Post-shock.- Immediately following vibration/shock, the periscope shall again be subjected to the test procedure specified in 4.6.3.1 and then to the tests in 4.6.3.5.1 and 4.6.3.5.2. Insure that the boresight knob scales are not disturbed from the original setting obtained in 4.6.3.1. Subsequent to the test in 4.6.3.5.1 and 4.6.3.5.2, the periscope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.4,3, prior to being subjected to the remaining tests specified herein. A visual examination shall also be made for missing or damaged parts, glass chips, foreign matter and bonding (separations, chipping or rupture). An audible examination shall be made for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a periscope defective.

4.6.3.5.1 Post-vibration/Post-shock collimation, passive reticle setting.- Observe through the eyepiece assembly (passive) and adjust by means of the boresight knobs to obtain coincidence of the periscope reticle boresight cross with the target image lines. Boresight knob adjustment shall not exceed the tolerance specified in 3.6.3.2, as measured on the boresight knob scales. Record collimation setting for use in the tests in 4.6.3.6.2.

4.6.3.5.2 Post-vibration/Post-shock collimation, daylight viewing area.- Perform this test in accordance with 4.6.3.1.1 Any difference from the recorded measurements established in 4.6.3.1.1 shall not exceed the tolerance specified in 3.6.3.2, as measured on target. Record collimation measurement for use in the test in 4.6.3.6.1.

4.6.3.6 Collimation repeatability.

4.6.3.6.1 Assembling daylight body to head assembly.- Perform this test subsequent to the test in 4.6.3.5.2. Remove the daylight body from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after each installation for repeatability of the daylight collimation measurement recorded in 4.6.3.5.2. The periscope shall meet the requirements of 3.6.4.1, as measured on the boresight knob scales.

4.6.3.6.2 Assembling passive elbow to head assembly.- Perform this test subsequent to the test in 4.6.3.5.1. Remove the passive elbow from the head assembly. Repeated removal and installation (a minimum of three times) shall be performed. The periscope shall be inspected after eanch installation for repeatability of the passive collimation setting recorded in 4.6.3.5.1. The periscope shall meet the requirements of 3.6.4.1, as measured on the boresight knob scales.

4.6.3.7 Modules.- Subsequent to the examinations and tests on the periscope, the head, body and elbow assemblies shall be separated and separately subjected to the "post-vibration" requirements and tests specified in their respective module specifications identified in 3.1. Subsequent to meeting

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the respective module specification post-vibration requirements and tests, the head, body and elbow assemblies shall be examined and tested in accordance with 4.6.3.8 to determine conformance to cleanliness as specified in 3.5.

4.6.3.8 Cleanliness.- Each separate assembly (see 4.6.4) shall be subjected to the tests in 4.6.3.8.1, 4.6.3.8.2 and 4.6.3.8.3 respectively, to determine compliance with 3.5.

4.6.3.8.1 Body assembly (daylight).- The optical surfaces of the body assembly shall be visually examined with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the eyepiece, inspection for dirt shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot candles. The body assembly (daylight) shall conform to the requirements of 3.5a and 3.5b(1).

4.6.3.8.2 Elbow assembly (passive).- The image intensifier tube shall be de-energized for this portion of the test. With a light source held close to the objective window and the eyelens, view the glass components of the elbow assembly to determine compliancw with the cleanliness requirements of 3.5a. Upon successful completion of this portion of the test, energize the image intensifier tube from a power source capable of supplying and maintaining 24 volts DC. The image tube shall be permitted a stabilization period of 15 seconds prior to performing the cleanliness test. Subsequent to the tube stabilization period, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the eyepiece diopter ring. The screen surface of the image tube shall then be visually inspected for cleanliness in accordance with 3.5b (2) and shall conform to the requirements specified therein. "Cold" and "Hot" spots on the screen surface of the image tube shall not be considered dirt.

4.6.3.8.3 Head assembly.- The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. While viewing through the exit window, inspection for dirt shall be made by viewing the illuminated field specified in 4.6.3.8.1. The head assembly shall conform to the requirements of 3.5a and 3.5b(3).

4.6.4 Performance.- To determine compliance with 3.6, the following tests shall be performed with the complete periscope positioned on the special test equipment specified in 4.5.2.3.1. A target projector collimator having a clear aperture of at least 2.0 inches and containing an appropriate calibrated horizontal and vertical reticle crossline target test pattern shall be utilized with the specialized test equipment specified herein. The vertical reticle target line shall be aligned plumb when in the test position on a qualified level surface and the target projector collimator shall be adjusted for infinity. The target projector collimator shall be positioned at the rear of the daylight body of the periscope so that its optical axis is in alignment with the line of sight (geometric center) of Lens 8289292. The optical axis of the target projector collimator shall be aligned parallel

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with respect to Datum Plane "J" and autocollimated within 10 seconds to Datum Plan "W". Unless otherwise specified during these tests, the Passive Focusing Ring 11733436 shall be adjusted for best focus of the target image after the diopter adjustment ring of the passive eyepiece is set for best focus of the phospher grain on the screen of the image tube; Eccentric 10516549 (for adjusting parallax) shall be adjusted for minimum parallax between the target image and reticle and then locked after the diopter adjustment knob of the daylight eyepiece is set for best focus of the projected infinity target; the clear filter shall be positioned in the field of view and the 10" collimator (8° field of view) shall be adjusted for 1200 meters. Light assembly "K" shall be removed when performing tests for daylight system. When performing tests for passive system, light assembly "K" shall be utilized. The image intensifier tube and reticle projector lamp shall be permitted a 15 second stabilization period, prior to conducting the performance tests.

4.6.4.1 Electrical characteristics - passive.

4.6.4.1.1 Image intensifier tube.- Apply 24 VDC ± 6 VDC to Contact 11733497 (Reference Drawing F11733401) and cycle the on-off switch from off to on, by means of the shutter lever. After allowing a 15 second tube stabilization period, view through the passive eyepiece and observe the image tube screen. The screen surface of the image intensifier tube shall be illuminated. Upon satisfactory completion of the test, remove the 24 VDC ± 6 VDC from Connector 11733497 and apply 3 VDC ± 0.3 VDC Connector 11733493 (Reference Drawing 11733401). Cycle the on-off switch from off to on. After allowing a 15 second tube stabilization period, view through the elbow assembly eyepiece. The screen surface of the tube shall be illuminated to determine overall compliance with 3.6.9.1.

CAUTION: When the on-off switch is in the "On" position and the input voltage to the regulator is interrupted, the tube cannot be energized by reapplication of the input voltage unless the on-off switch is cycled from off to on.

4.6.4.1.2 Image steadiness.- Perform this test with the reticle projector lamp off. Positon an observation telescope at the passive eyepiece of the periscope. The telescope shall contain a crossline reticle pattern and be capable of focusing on the screen of the image tube. With the aid of the telescope, view through the passive eyepeice and establish the telescope reticle crosslines coincident to the projected target image crosslines. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker, no movement of the image nor audible sparking as specified in 3.6.9.2. Image movement shall be determined by means of the target image with respect to the observation telescope reticle pattern. Sparking shall be determined by means of an audio examination. Upon completion of this test, energize the reticle projector lamp.

4.6.4.1.3 Reticle brightness.- Perform this test concurrent with the test in 4.6.4.1.1. With each voltage specified in 3.6.9.3 applied to Contact 11733497 (Reference Drawing F11733401) and the image intensifier tube energized, rotate the reticle brightness control knob MS 91528-0E1B to the off position. Then while viewing through the elbow assembly eyepiece, rotate the control knob clockwise throughout its range of movement. During

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this clockwise rotation of the control knob, the reticle brightness shall be adjustable from off to full bright and the reticle markings shall appear clearly defined at each voltage setting when observed through the elbow assembly eyepiece to determine compliance with 3.6.9.3.

4.6.4.1.4 On-off switch.- This test may be performed at the same time the test of 4.6.4.1.1 is performed. With the voltage specified in 4.6.4.1.1 applied to Contact 11733497, view through the elbow assembly eyepiece and observe the illuminated screen surface of the tube. Then while the on-off switch is in the on position, remove the voltage from the contact and reapply. During the above observation, the screen surface of the tube shall shut off when voltage is removed and shall not illuminate by reapplication of the voltage unless the on-off switch is recycled from off to on to be in conformance with 3.6.9.4.

4.6.4.2 Parallax.- View through the daylight eyepiece and observe the target (1200 meters) image crosslines and the projected infinity target image crosslines. While viewing through the eyepiece, parallax will be recognized by any apparent displacement of the intersecting point of the target (1200 meters) crosslines with respect to the intersecting point of the projected infinity target image crosslines when the observer's head is moved from side to side or up and down. Displacement between the intersecting points of the two target images shall not exceed the tolerance specified in 3.6.2. Upon satisfactory completion of this test, perform the tests specified in 4.6.4.7.1 and 4.6.4.2.1.

4.6.4.2.1 Parallax adjustment.- If adjustment is necessary, a standard type screw driver that properly fits the screw slot of the locking screw on Clamp 10516556 and Eccentric 10516549 shall be utilized for performing this adjustment. Back off the locking screw a minimum of two revolutions from its locked position. Then while performing the test in 4.6.4.2, rotate the eccentric clockwise or counterclockwise until the parallax requirement specified therein has been met. When the adjustment is complete, apply the applicable torque value specified in 3.6.10.8 to the locking screw utilizing the torque wrench specified in 4.6.2.8.

4.6.4.3 Passive viewing area.

4.6.4.3.1 Elevation knob travel.- Establish coincidence of the periscope reticle boresight cross and the target (1200 meters) image crosslines by means of the boresight knobs. While observing the boresight cross (intersecting point), rotate the elevation knob in the clockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a as measured on the target. Re-establish coincidence of the reticle boresight cross and the target image crosslines by means of the boresight knobs. While observing the boresight cross, rotate the elevation knob in the counterclockwise direction. The boresight cross shall travel as specified in 3.6.3.1.2a as measured on the target.

4.6.4.3.2 Deflection knob travel.- Perform this test utilizing the procedure in 4.6.4.3.1 except, the deflection knob shall be rotated in the clockwise and counterclockwise direction. Boresight cross (intersecting point) travel shall meet the requirements specified in 3.6.3.1.2b as measured on the target.

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4.6.4.3.3 Line of sight variation.- Establish coincidence of the periscope reticle boresight cross and the target (1200 meters) image crosslines by means of the boresight knobs. While observing the intersecting point of the reticle boresight cross, rotate the deflection knob to its clockwise stop. During this rotation of the deflection knob any variation in elevation in the line of sight through the periscope reticle boresight cross, shall not exceed the values specified in 3.6.3.1.2c. Re-establish coincidence of the periscope reticle boresight cross and the target image crosslines by means of the boresight knobs. While observing the intersecting point of the periscope reticle boresight cross, rotat the deflection knob to its counterclockwise stop. During this rotation of the deflection knob, any variation in elevation in the line of sight through the periscope reticle boresight cross shall not exceed the value specified in 3.6.3.1.2c to determine overall conformance with the requirements specified therein.

4.6.4.4 Image tilt - daylight and passive.- The daylight and passive optical systems of the periscope shall be inspected utilizing holder "T" containing telescope "J" with special reticle "L" (Ref. Dwg. F11738140). The vertical reticle line of the telescope shall be aligned plumb when in the test position on a qualified level surface. With the aid of the auxiliary telescope, view through the passive eyepiece of the periscope and establish the telescopes reticle crosslines coincident to the target (1200 meters) image crosslines. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.6.5, as determined by the tolerance gates provided on the telescope reticle. Repeat the above procedure with telescope positioned at the daylight eyepiece of the periscope except, the telescopes reticle crosslines shall be established coincident to the infinity target image crosslines. The vertical reticle line of the infinity target image shall be parallel to the Telescope vertical reticle line within the tolerance specified in 3.6.5, as determined by the tolerance gates provided on the telescope reticle.

4.6.4.5 Parallelism, of reticle and image, passive.- Perform this test while viewing through the passive eyepiece of the periscope with the unaided eye. Position one extremity of the periscope horizontal reticle line coincident to the horizontal target (1200 meters) image line. The periscope horizontal reticle line shall be parallel to the horizontal target line within the tolerance specified in 3.6.6, as measured on the target.

4.6.4.6 Parallelism of target image and projected image.- Perform this test while viewing through the daylight eyepiece of the periscope with the unaided eye. Observe the horizontal line of the target (1200 meters) image with respect to the horizontal line of projected infinity target image. The horizontal lines shall be parallel within the tolerance specified in 3.6.6.1, as determined by the tolerance gates provided on the target.

4.6.4.7 Objective focusing range.

4.6.4.7.1 Daylight.- This test shall be performed subsequent to the test in 4.6.4.2. A dial indicator, graduated in .001 of an inch, mounted

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of an appropriate stand, a supporting medium capable of receiving the dial indicator as well as the body assembly to be tested and the screw driver specified in 4.6.4.2.1 shall be utilized for this test. With the daylight body removed from the periscope and firmly secured on the supporting medium, position the dial indicator so that its contact point touches the face of the movable objective tube of the daylight body and set the dial to read zero with the indicating hand. Back off the locking screw of Clamp 10516556 a minimum of two full revolutions from its locked position. Starting from the zero reading of the dial indicator, rotate the eccentric until the objective assembly has traveled the minimum distance specified in 3.6.7.1, as measured on the dial indicator. Return the indicating hand of the dial indicator to zero by means of the eccentric and repeat this travel in the opposite direction. There shall be sufficient travel in each direction to meet the minimum requirements specified in 3.6.7.1, as measured on the dial indicator. Upon completion of this test, reinstall the dayligh body in periscope and repeat the parallax adjustment procedure specified in 4.6.4.2.1. The daylight body shall then be subjected to the test in 4.6.2.8.

4.6.4.7.2 Passive.- This portion of the test shall be performed with the 10" collimator adjusted for 50 meters. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, rotate focusing ring of the elbow assembly for best focus of target image to determine compliance with 3.6.7.2. Upon satisfactory completion of this portion of the test, the 10" collimator shall be readjusted for infinity to determine compliance with the 3,000 meters specified in 3.6.7.2. With the diopter ring adjusted for best focus of the phosphor grain on the screen of the image tube, again rotate the focusing ring of the elbow assembly for best focus of the target image to determine overall compliance with 3.6.7.2.

4.6.4.8 Resolution - daylight.- The daylight optical system of the periscope shall be inspected for resolution utilizing a calibrated dioptometer with a manification of at least three power. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter adjustment knob of the periscope daylight eyepiece to indicate zero diopter. Position the dioptometer at the periscope daylight eyepiece. While viewing through the dioptometer, adjust the objective scale of the dioptometer for best focus of the resolution pattern appearing in the fourth quadrant of the target reticle located in the 10" Collimator (Ref. Dwg. F11738140 Test Set-up, Periscope). Any change in diopter readings by adjustment of the dioptometer objective scale to obtain the detectable line structure and proper line count in all four meridians shall be recorded, as measured on the dioptometer objective scale. The difference between the recorded readings for any two meridians shall not exceed the value specified in 3.6.8.1.

4.6.4.9 Resolution - passive.- This test shall be performed while viewing through the passive eyepiece with the unaided eye. Best focus of both meridians of resolution pattern appearing in the first quadrant of the target reticle located in the 10" Collimator (Ref. Dwg. F11738140 Test Set-Up Periscope), shall be obtained by adjusting the passive eyepiece's diopter ring. Any change in the diopter readings by adjustment of the passive eyepiece diopter ring to obtain the detectable line structure and

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proper line count in each meridian shall be recorded, as measured on the ring scale. The difference between the readings for the meridians shall not exceed the value specified in 3.6.8.2.

4.6.5 Interchangeability.

4.6.5.1 Modules.- Each sample periscope (see 4.4.1.1) shall be disassembled (removal of passive elbow from head and daylight body from head) to permit interchange of the component modules. The module shall be identified and interchanged as shown in Table VI.

TABLE VI

Original module positions

	PERISCOPE 1	Periscope 2	Periscope 3
Head Assembly (HA	HA-1	HA-2	HA-2
Body Assembly (DL) (Daylight)	DL-1	DL-2	DL-3
Elbow Assembly (P) (Passive)	P-1	P-2	P-3

Interchanged Module Positions

	Periscope 1	Periscope 2	Periscope 3
Head Assembly (HA)	HA-1	HA-2	HA-3
Body Assembly (DL) (Daylight)	DL-2	DL-3	DL-1
Body Assembly (P) (Passive)	P-3	P-1	P-2

Upon reassembly of the three interchanged periscopes, each periscope shall be subjected to the tests specified in 4.6.4.2, 4.6.4.4, 4.6.4.5, 4.6.4.6, 4.6.4.8 and 4.6.4.9 to determine compliance with 3.2a.

4.6.5.2 Periscope with mount.- This test shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings F11732156 and F11732155 respectively, to determine compliance with 3.2b.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11727450. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This periscope is utilized as a day or night sighting and observation instrument. This capability is made posible by the use of a passive sighting system and a conventional periscope optical system encased in a single body assembly. The passive portion utilized the 24 volt vehicle current source which is converted to a 2.7

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VDC required voltage to actuate the image intensifier tube (passive).

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable package data sheet (see 5.1)
d. Applicable stock number
e. Provisions for First Article Testing

6.3 Input coupling.- At no time shall the input coupling be allowed to strike hard against the travel stops.

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